



KNACK ENGINEERING SERVICES PVT. LTD.



Test Report no:	KES260330236	Report Issue Date:	02./07/2026	ULR No.	TC1525026000001330F
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Customer Party Name: AIGIS LOGISTICS LIMITED

Project / Site Address: Proposed Construction of Tank at Pirpau, Mahul. (T-408)

Test Start Date: 31/03/2026	Test End Date :31/03/2026
Pile ID: T409-P14	
Size:800 mm	Grade: M-30
Main Client: AIGIS Logistics Limited	Consultant Name: -
EPC Contractor: -	
Discipline: MECHANICAL	Group: Soil-Field - MECHANICAL
Test Name: Lateral Load Test	
References: IS 2911 Part 4	



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1. GENERAL PHILOSOPHY

One (01) No. of Routine Pile Lateral Load Test was conducted to reconfirm the Lateral Load Carrying Capacity of the Piles to support the various design lateral loads due to Wind or Earthquake etc. or any other lateral load on the structure. This report provides the details for Load – Displacement behaviour of this Test pile.

2. METHODOLOGY FOR LATERAL PILE LOAD TESTING:

2.1 Scope of Work:

This procedure was applicable for the testing of 1200mm diameter bored Initial Test pile for the Lateral Load.

1. Mobilization of required manpower & machineries.
2. Setting up of testing apparatus.
3. Incremental loading and unloading & recording of Load Vs Displacement.
4. Preparation of test report.

2.2 Reference:

- IS 2911 (Part 4) - 2013 -Code of practice for design and construction of pile foundations- Part 4 Load Test on Piles.

2.3 Equipments Used:

- Hydraulic Jack – 100 MT Capacity
- Pressure Gauge – 0-280 kg/cm² with LC of 5.0 kg/cm²
- Dial Gauges – 2 Nos of Dial Gauges of 25 mm travel with LC of 0.01 mm

2.4 Setting up of Testing Apparatus:

- 1) The test Apparatus used for the test consisted of the following:
 - i. Hydraulic Jack – 100 MT Capacity
 - ii. Pressure Gauge – 0 – 280 kg/cm² range
 - iii. Dial Gauges – 2 Nos. with LC of 0.01 mm
 - iv. Datum Bars – 1 Nos.
 - v. Suitable Beam with End Plates for Transfer of Load from jack to reaction Pile.
- 2) The liner / concrete of test pile was levelled for 100 mm x 100 mm or 100 mm dia. to fix the hydraulic Jacks. The reaction was taken from nearest Initial Test Pile (min 2.00m distance apart) The hydraulic jack was resting on steel stool & Datum Bar was erected.

2.5 Load Test Procedure:

- 1) The test was carried out by applying a series of lateral incremental load as per cl. 8.1 of IS 2911 (IV) 2013 each increment being of about 20 percent (20%) of ultimate load on the pile. This increment was carried out up to 150% of the safe working load as per cl. 8.4.1 of IS 2911 (IV) 2013.
- 2) Displacement was recorded with 2 dial gauges spaced at 30 cm and kept horizontally one above the other on test pile (preferable one dial gauge 15cm above and the other 15cm below the cut off level).
- 3) The next increment was applied after the rate of displacement is less than or equal to 0.1 mm per 30 min subject to minimum of 30 min as per cl. 8.2 of IS 2911 (IV) 2013.
- 4) After completion of loading up to final load was decreased, each decrement being about 20 % of safe load on pile after every 1,10,20 and 30 minutes interval. Readings were taken immediately after unloading and 15 minutes later, for each stage.
- 5) All of the above activities were carried out in presence of Engineer's Representative.
- 6) Once the Test Load was reached, the safe Lateral load on single pile for the Routine test would be least of the following:
 - i) Fifty percent of the final load at which the total displacement increases to 12 mm;
 - ii) Final load at which the total displacement corresponds to 5 mm.



2.6 Preparation of Test Report:

A chart was plotted to depict Lateral Load on Pile in Tons vs. Pile Deflection in mm and same is attached with the test report.

2.7 Conclusion:

As can be seen from the results of Lateral Load tests on Working Test Pile No. T409-P14, the Deflection of the piles at the Test Load is 7.87 mm. Moreover, Net Deflection of Pile is 5.09 & Elastic Rebound is 2.98 mm.

Hence, as the total Displacement of the pile is more than 5mm, the Design Loads can be considered as Safe Design Lateral Loads for the piles.

Safe Design Lateral Load for pile T-409-P14 = 5.80 MT

FOR KNACK ENGINEERING SERVICES PRIVATE LIMITED,

Prepared By
Sohel Shaikh

Sr. Site Engineer

Checked By
Ashfaq Ansari

LAB-HOD
(Authorized Signatory)



Annexure I

(TEST READINGS)

KNACK ENGINEERING SERVICES PRIVATE LIMITED			
LATERAL PILE LOAD TEST (IS 2911: Part4 2013)			
Client Name: ARUNIS	FORMAT NO. KES/FM/50/DS-LPLT		
Project Name: oil storage tank	Date of Testing:		
Consultants:	Sample ID:		
Name:	Site Location: T-409-P14		
Contractor:	Grade of Concrete: M-30		
Name:	Contact Person:		
Test Load in MT: 14 MT	Start Time: 8:00 PM	End Time: 8:20 PM	

TIME(Minutes)	PRESSURE (Kg/cm ²)	LOAD (MT)	DG 1 READING (mm)	DG 2 READING (mm)	DG 3 READING (mm)	AVERAGE READING (mm)	REMARKS
0			0.07	0.35			
15	20		1.31	1.35			
30			1.32	1.58			
0			1.32	1.58			
15	40		3.72	3.75			
30			3.73	3.76			
0			3.73	3.76			
15	60		5.46	5.80			
30			5.42	5.82			
0			5.42	5.82			
15	80		6.95	6.34			
30			6.97	6.35			
0			6.97	6.35			
15	95		8.01	8.04			
30			8.03	8.12			

Tested By (Name & Sign):

Sohel Shalikh

PMC (Name & Sign)

Witness by (Name & Sign)

UNLOADING

TIME(Minutes)	PRESSURE (Kg/cm ²)	LOAD (MT)	DG 1 READING (mm)	DG 2 READING (mm)	DG 3 READING (mm)	DG 4 READING (mm)	AVERAGE READING (mm)	REMARKS
2	80		7.95	7.98				
2	60		7.65	7.72				
2	40		6.90	6.91				
2	20		6.20	6.21				
2	0		5.50	5.10				

Tested By (Name & Sign):

Sohel Shalikh

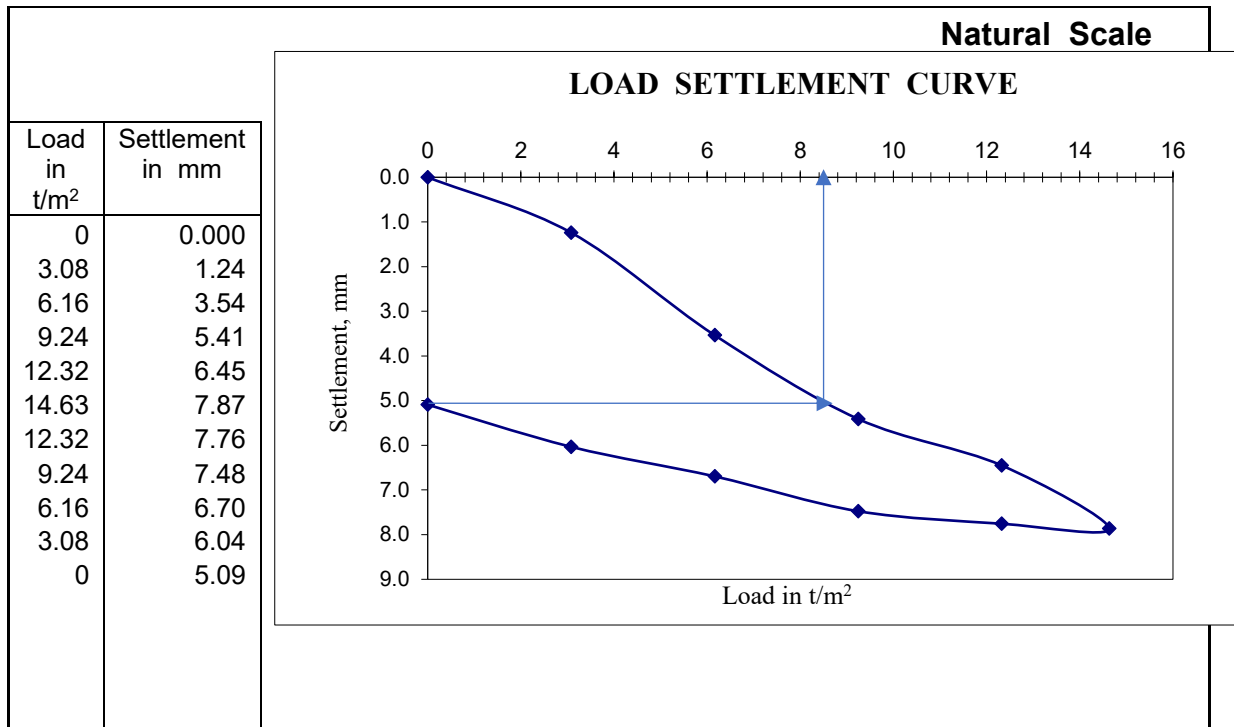
PMC (Name & Sign)

Witness by (Name & Sign)



Annexure II

(LOAD VS. DEFLECTION PLOT)





Annexure III

(SITE PHOTOGRAPHS)



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*****END OF REPORT*****